

Predictive Analysis Using Machine Learning

1. Introduction

Predictive analysis is a branch of machine learning that focuses on predicting future outcomes based on historical data. In this project, we build a classification model to predict whether a person will purchase a product based on age and salary.

2. Objective

The main objective of this project is to develop a machine learning model that can accurately predict customer purchase behavior. We use classification algorithms such as Logistic Regression and Decision Tree to achieve this goal.

3. Dataset Description

The dataset consists of three features: Age, Salary, and Purchased. Age and Salary are independent variables, while Purchased is the dependent variable with binary values (0 = Not Purchased, 1 = Purchased).

4. Data Visualization

A scatter plot is used to visualize the relationship between Age and Salary. The color of each point represents whether the product was purchased or not. This helps in understanding patterns and separability in the data.

5. Data Preprocessing

The dataset is split into training and testing sets using an 80-20 ratio. Feature scaling is applied using StandardScaler to normalize the data, which improves the performance of machine learning models.

6. Model Training

Two models are used in this project: Logistic Regression and Decision Tree Classifier. Logistic Regression is effective for binary classification, while Decision Tree can capture non-linear patterns in the data.

7. Model Evaluation

The performance of the models is evaluated using accuracy score, confusion matrix, and classification report. These metrics provide insights into model performance and prediction quality.

8. Results and Comparison

Both models were successfully trained and tested. Logistic Regression provided stable and consistent results, while Decision Tree showed flexibility in capturing patterns. The accuracy scores help compare both models.

9. Conclusion

This project demonstrates how machine learning can be used for predictive analysis. It highlights the importance of preprocessing, model selection, and evaluation in building effective predictive systems.